Response to Reviewers — Second Cut

Team 3 · *Servo Characterization for the SOFIA Modular Kinetic Surface*

Acknowledgements

The authors thank the reviewers for their careful evaluation of the first-cut manuscript. <One sentence on how the comments improved the paper.> All comments were considered; the responses below follow the keys of the review report (R1.1, R1.2, . . .). Line, section, figure, table, and equation numbers refer to the revised manuscript that accompanies this letter.

Main changes. (i) <Main change 1> (Section **??**). (ii) <Main change 2>. (iii) <Main change 3>.

Summary of changes (observation–response–change matrix)

6 observations: 5 addressed, 0 partially addressed, 1 not addressed (justified).

Key	Status	Location in the revised paper
R1.1	**[Addressed]**	Lines **??**–**??**
R1.2	**[Addressed]**	<Ubicacion>
R2.4	**[Addressed]**	lines 36-47
R3.1	**[Addressed]**	Fig. 1
R3.3	**[Addressed]**	lines 193-199
R4.1	**[Not addressed (justified)]**	Lines **??**–**??**

Reviewer 1 — *Format, LaTeX and template compliance*

R1.1 [Addressed]

Reviewer comment: <Comentario del revisor, copiado textual del PDF de observaciones.>

Response: As suggested, we <describe what was changed> (lines **??**–**??**).

Changes in the revised paper: Lines **??**–**??**

R1.2 [Addressed]

Reviewer comment: <Comentario textual.>

Response: <Response.>

Changes in the revised paper: <Ubicacion>

Reviewer 2 — *Writing and structure*

R2.4 [Addressed]

Reviewer comment: Salvo un párrafo final, Materials and Methods está formada solo por tablas; las subsecciones A–F no tienen texto que explique qué muestra cada tabla ni qué se concluye.

Response: We rewrote the introduction and all mentions of servo characterization to explicitly state that it is the proposed contribution of this work. We simplified the work to only have the servo characterization as our only differentiation. (lines 36-47). The revised text now reads:

> *"This work proposes an experimental method to estimate individual servo time constraints with a mechanical solution"*

Changes in the revised paper: lines 36-47

Reviewer 3 — *Technical content*

R3.1 [Addressed]

Reviewer comment: No hay arquitectura preliminar del sistema en figura: cómo se conectan la cámara y el PC, la ESP32-C3, los dos PCA9685, los 32 servos, los transceptores RS-485 y la alimentación

Response: Added a System Architecture Diagram Fig. 1 showing the host, ESP32-C3, dual PCA9685 drivers, 32 servos, RS-485 transceivers.

Changes in the revised paper: Fig. 1

R3.3 [Addressed]

Reviewer comment: La "constante de máquina" o "mechanical time constant" del servo no se define: no hay modelo, variables ni ecuaciones. El diferenciador promete compararla contra el "valor teórico" del MG90S, que la hoja de datos no reporta.

Response: Formulated the first-order dynamic model $\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m}$ and clarified that the reference baseline is derived from the manufacturer's no-load speed specification (0.10 s/60° at 4.8 V).

Changes in the revised paper: lines 193-199

Reviewer 4 — *D-Lab and state of the art*

R4.1 [Not addressed (justified)]

Reviewer comment: <Comentario textual.>

Response: We thank the reviewer for this suggestion, and we agree that <...>. However, <technical reason: scope, cost, time, available evidence>. Instead, <what was done>, as stated in lines **??–??**.

Changes in the revised paper: Lines **??–??**